

Day 3 Of Communication System Class

1.1 Review of signals and systems

2.1 Analog and digital data

2.2 Analog signals, periodic and aperiodic signals, periodic signals characteristics (Time, frequency domain)

2.3 Introduction to Fourier series representation of periodic signal, Fourier transform representation

2 Representation of signals and systems in communication

2.1 Review of signals (types, mathematical representation and applications)

2.2 Linear/non-linear, time variant/invariant systems. Impulse response and transfer function of a system. Properties of LTI systems.

2.3 Low pass and band pass signals and systems, bandwidth of the system, distortionless transmission, the Hilbert transform and its properties.

2.4 Complex envelopes rectangular (in-phase and quadrature components) and polar representation of band pass band limited signals

3 Spectral Analysis

3.1 Review of Fourier series and transform, energy and power, Parseval's theorem

3.2 Energy Density Spectrum, periodogram, power spectral density function (psdf)

3.3 Power spectral density functions of harmonic signal and white noise

3.4 The autocorrelation (AC) function, relationship between psdf and AC function

Introduction to Signals and Systems

1.1 Signal

Any physical phenomenon that conveys or carries some information can be called a signal. The music, speech, motion pictures, still photos, heart beat, etc., are examples of signals that we normally encounter in day to day life.

Usually, the information carried by a signal will be a function of an independent variable. The independent variable can be time, spatial coordinates, intensity of colours, pressure, temperature, etc.,. The most popular independent variable in signals is time and it is represented by the letter “t”.

The value of a signal at any specified value of the independent variable is called its **amplitude**. The sketch or plot of the amplitude of a signal as a function of independent variable is called its **waveform**.

Mathematically, any signal can be represented as a function of one or more independent variables. Therefore, a **signal** is defined as any physical quantity that varies with one or more independent variables.

1.1.1 Continuous Time Signal

In a signal with time as independent variable, if the signal is defined continuously for any value of the independent variable time “t”, then the signal is called **continuous time signal**. The continuous time signal is denoted as “x(t)”.

The continuous time signal is defined for every instant of the independent variable time and so the magnitude (or the value) of continuous time signal is continuous in the specified range of time. Here both the magnitude of the signal and the independent variable are continuous.

1.1.2 Discrete Time Signal

In a signal with time as independent variable, if the signal is defined only for discrete instants of the independent variable time, then the signal is called **discrete time signal**.

1.2 System

Any process that exhibits cause and effect relation can be called a **system**. A system will have an input signal and an output signal. The output signal will be a processed version of the input signal. A system is either interconnection of hardware devices or software / algorithm.

A system is denoted by letter $\mathcal{H}$. The diagrammatic representation of a system is shown in fig 1.1.

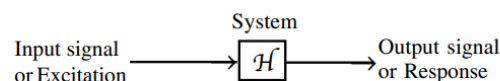


Fig 1.1 : Representation of a system.

The operation performed by a system on input signal to produce output signal can be expressed as,

$$\text{Output} = \mathcal{H}\{\text{Input}\}$$

where $\mathcal{H}$ denotes the system operation (also called **system operator**).

When a continuous time system satisfies the properties of linearity and time invariance then it is called LTI (Linear Time Invariant) continuous time system. Most of the practical systems that we encounter in science and engineering are LTI systems.

A system which can process discrete time signal is called discrete time system, and so the input and output signals of a discrete time system are discrete time signals.

A system which can process continuous time signal is called continuous time system, and so the input and output signals of a continuous time system are continuous time signal.

Continuous Time Signals and Systems

2.2 Standard Continuous Time Signals

1. Impulse signal

The impulse signal is a signal with infinite magnitude and zero duration, but with an area of A . Mathematically, impulse signal is defined as,

$$\text{Impulse Signal, } \delta(t) = \infty ; t = 0 \quad \text{and} \quad \int_{-\infty}^{+\infty} \delta(t) dt = A \\ = 0 ; t \neq 0$$

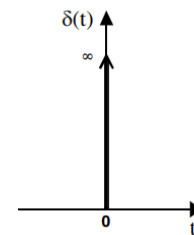


Fig 2.1 : Impulse signal (or Unit Impulse signal).

The unit impulse signal is a signal with infinite magnitude and zero duration, but with unit area. Mathematically, unit impulse signal is defined as,

$$\text{Unit Impulse Signal, } \delta(t) = \infty ; t = 0 \quad \text{and} \quad \int_{-\infty}^{+\infty} \delta(t) dt = 1 \\ = 0 ; t \neq 0$$

2. Step signal

The step signal is defined as,

$$x(t) = A ; t \geq 0 \\ = 0 ; t < 0$$

The unit step signal is defined as,

$$x(t) = u(t) = 1 ; t \geq 0 \\ = 0 ; t < 0$$

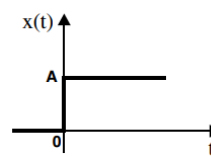


Fig 2.2 : Step signal.

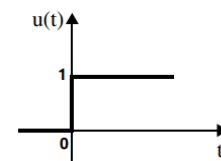


Fig 2.3 : Unit step signal.

3. Ramp signal

The ramp signal is defined as,

$$x(t) = At ; t \geq 0 \\ = 0 ; t < 0$$

The unit ramp signal is defined as,

$$x(t) = t ; t \geq 0 \\ = 0 ; t < 0$$

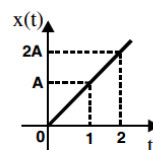


Fig 2.4 : Ramp signal.

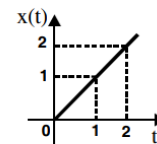


Fig 2.5 : Unit ramp signal.

4. Parabolic signal

The parabolic signal is defined as,

$$x(t) = \frac{At^2}{2} ; \text{ for } t \geq 0$$

$$= 0 ; \quad t < 0$$

The unit parabolic signal is defined as,

$$x(t) = \frac{t^2}{2} ; \text{ for } t \geq 0$$

$$= 0 ; \quad t < 0$$

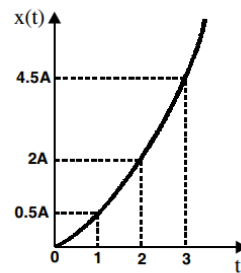


Fig 2.6 : Parabolic signal.

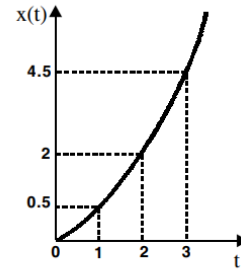


Fig 2.7 : Unit parabolic signal.

5. Unit pulse signal

The unit pulse signal is defined as,

$$x(t) = \Pi(t) = u\left(t + \frac{1}{2}\right) - u\left(t - \frac{1}{2}\right)$$

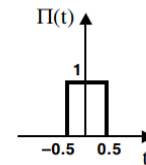


Fig 2.8 : Unit pulse signal.

10. Signum signal

The Signum signal is defined as the sign of the independent variable t . Therefore, the Signum signal is expressed as,

$$x(t) = \text{sgn}(t) = 1 ; t > 0$$

$$= 0 ; t = 0$$

$$= -1 ; t < 0$$

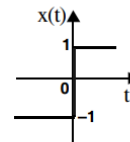


Fig 2.16 : Signum signal.

11. Sinc signal

The Sinc signal is defined as,

$$x(t) = \text{sinc}(t) = \frac{\sin t}{t} ; -\infty < t < \infty$$

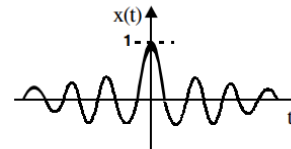


Fig 2.17 : Sinc signal.

2.3 Classification of Continuous Time Signals

The continuous time signals are classified depending on their characteristics. Some ways of classifying continuous time signals are,

1. Deterministic and Nondeterministic signals
2. Periodic and Nonperiodic signals
3. Symmetric and Antisymmetric signals (Even and Odd signals)
4. Energy and Power signals
5. Causal and Noncausal signals

2.3.1 Deterministic and Nondeterministic Signals

The signal that can be completely specified by a mathematical equation is called a **deterministic signal**. The step, ramp, exponential and sinusoidal signals are examples of deterministic signals.

Examples of deterministic signals: $x_1(t) = At$ $x_2(t) = X_m \sin \Omega_0 t$
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The signal whose characteristics are random in nature is called a **nondeterministic signal**. The noise signals from various sources like electronic amplifiers, oscillators, radio receivers, etc., are best examples of nondeterministic signals.

2.3.2 Periodic and Nonperiodic Signals

A periodic signal will have a definite pattern that repeats again and again over a certain period of time. Therefore the signal which satisfies the condition,

$$x(t + T) = x(t) \text{ is called a } \textbf{periodic signal}.$$

A signal which does not satisfy the condition, $x(t + T) = x(t)$ is called an **aperiodic or nonperiodic signal**. In periodic signals, the term T is called the **fundamental time period** of the signal. Hence, inverse of T is called the **fundamental frequency**, F_0 in cycles/sec or Hz, and $2\pi F_0 = \Omega_0$ is called the **fundamental angular frequency** in rad/sec.

The sinusoidal signals and complex exponential signals are always periodic with a periodicity of T , where, $T = \frac{1}{F_0} = \frac{2\pi}{\Omega_0}$. The proof of this concept is given below.

2.3.3 Symmetric (Even) and Antisymmetric (Odd) Signals

The signals may exhibit symmetry or antisymmetry with respect to $t = 0$.

When a signal exhibits symmetry with respect to $t = 0$ then it is called an **even signal**. Therefore, the even signal satisfies the condition, $x(-t) = x(t)$.

When a signal exhibits antisymmetry with respect to $t = 0$, then it is called an **odd signal**. Therefore, the odd signal satisfies the condition, $x(-t) = -x(t)$.

Since $\cos(-\theta) = \cos\theta$, the cosinusoidal signals are even signals and since $\sin(-\theta) = -\sin\theta$, the sinusoidal signals are odd signals.

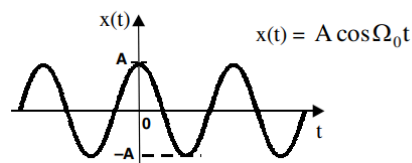


Fig 2.19a : Symmetric or Even signal.

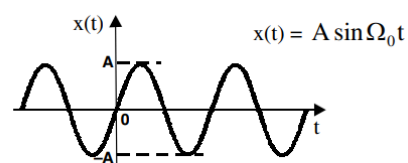


Fig 2.19b : Antisymmetric or Odd signal.

Fig 2.19 : Symmetric and antisymmetric continuous time signals.

A continuous time signal $x(t)$ which is neither even nor odd can be expressed as a sum of even and odd signal.

$$\text{Let, } x(t) = x_e(t) + x_o(t)$$

where, $x_e(t)$ = Even part of $x(t)$ and $x_o(t)$ = Odd part of $x(t)$

Now, it can be proved that,

$$\begin{aligned} x_e(t) &= \frac{1}{2}[x(t) + x(-t)] \\ x_o(t) &= \frac{1}{2}[x(t) - x(-t)] \end{aligned}$$

2.3.4 Energy and Power Signals

The signals which have finite energy are called **energy signals**. The nonperiodic signals like exponential signals will have constant energy and so nonperiodic signals are energy signals.

The signals which have finite average power are called **power signals**. The periodic signals like sinusoidal and complex exponential signals will have constant power and so periodic signals are power signals.

The **energy** E of a continuous time signal $x(t)$ is defined as,

$$\text{Energy, } E = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt \text{ in joules}$$

The average **power** of a continuous time signal $x(t)$ is defined as,

$$\text{Power, } P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \text{ in watts}$$

For periodic signals, the average power over one period will be same as average power over an infinite interval.

$$\therefore \text{ For periodic signals, power, } P = \frac{1}{T} \int_0^T |x(t)|^2 dt$$

For energy signals, the energy will be finite (or constant) and average power will be zero. For power signals the average power is finite (or constant) and energy will be infinite.

i.e., For energy signal, E is constant (i.e., $0 < E < \infty$) and $P = 0$.

For power signal, P is constant (i.e., $0 < P < \infty$) and $E = \infty$.

Example 2.4

Determine the power and energy for the following continuous time signals.

a) $x(t) = e^{-2t} u(t)$

b) $x(t) = e^{j\left(2t + \frac{\pi}{4}\right)}$

c) $x(t) = 3\cos 5\Omega_0 t$

2.3.5 Causal, Noncausal and Anticausal Signals

A signal is said to be **causal**, if it is defined for $t \geq 0$.

Therefore if $x(t)$ is causal, then $x(t) = 0$, for $t < 0$.

A signal is said to be **noncausal**, if it is defined for either $t \leq 0$, or for both $t \leq 0$ and $t > 0$.

Therefore if $x(t)$ is noncausal, then $x(t) \neq 0$, for $t < 0$.

When a noncausal signal is defined only for $t \leq 0$, it is called **anticausal signal**.

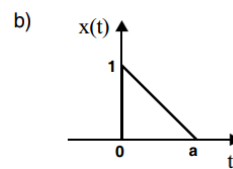
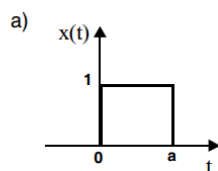
Examples of causal and noncausal signals

Step signal,	$x(t) = A$	$t \geq 0$	} Causal signals
Unit step signal,	$x(t) = u(t) = 1$	$t \geq 0$	
Exponential signal,	$x(t) = A e^{bt} u(t)$		
Complex exponential signal,	$x(t) = A e^{j\Omega_0 t} u(t)$		
Exponential signal,	$x(t) = A e^{bt}$	for all t	} Noncausal signals
Complex exponential signal,	$x(t) = A e^{j\Omega_0 t}$	for all t	

Note : On multiplying a noncausal signal by $u(t)$, it becomes causal.

Example 2.6

Sketch the even and odd parts of the following signals.

**Solution**

a) The even part of the signal is given by, $x_e(t) = \frac{1}{2}[x(t) + x(-t)] = \frac{1}{2} x(t) + \frac{1}{2} x(-t)$ (1)

The odd part of the signal is given by, $x_o(t) = \frac{1}{2}[x(t) - x(-t)] = \frac{1}{2} x(t) - \frac{1}{2} x(-t)$ (2)